Curriculum Vitae

Abdesselam FERDI, PhD

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Date of the CV: September 10, 2026

RESEARCH PROFILE

Postdoctoral researcher focused on the development of efficient and interpretable AI methods for clinical and real-world applications.

KEY CONTRIBUTIONS

- **Interpretable image classification with foundation models:** Developed interpretable machine learning models for image classification based on LLMs and VLMs (under review, 2026).
- **Physics-inspired model pruning:** Developed Electrostatic Force and Gravity regularization methods for efficient neural networks (*Information Sciences*, 2026; *IEEE ICMLANT*, 2024).
- **Lightweight machine learning models:** Designed lightweight architectures (G-YOLO and RED-CNN) for fracture detection and LDCT denoising (*Biomedical Signal Processing and Control*, 2026; *Multimedia Tools and Applications*, 2025).

RESEARCH INTERESTS

Efficient and interpretable AI • Physics-inspired machine learning • Medical AI

PROFESSIONAL EXPERIENCE

Research Fellow (Explainable AI) 2025–present
Institute of Computer Science, University of Tartu, Estonia
Link: https://www.etis.ee/CV/Abdesselam_Ferdi
Research: Development of novel interpretable-by-design machine learning models for image classification based on foundation models.
Teaching: Teach a full seminar course on Explainable AI to undergraduate and graduate students.

Researcher (Model compression) 2023–2024
Laboratoire LISSI, Université Paris-Est Créteil, Paris, France
Link: https://www.lissi.fr/home/

Research: (1) Development of advanced deep learning algorithms for the denoising of low-dose CT images; (2) Development of two novel pruning methods for the acceleration of deep learning algorithms.
Teaching: Taught courses on deep learning and Python programming to undergraduate and graduate students.

Teaching assistant 2021–2023
Constantine 1 University & ENS Biotechnology, Constantine, Algeria
Links: https://www.umc.edu.dz/index.php/en and https://wp.ensbiotech.edu.dz/en/
Teaching: Taught courses on image processing, electronics, numerical methods, informatics, and Matlab programming to undergraduate and graduate students.

Biomedical Engineering Intern 2018
Dr. Benbadis University Hospital Center, Constantine, Algeria
Role: Participated in the maintenance of biomedical equipment and hospital utilities and collaborated with engineers on technical repairs and preventive maintenance protocols.

EDUCATION

PhD in Biomedical Engineering 2021–2024
SP-LAB, Electronics Department, Constantine 1 University, Constantine, Algeria
Link: https://www.umc.edu.dz/index.php/en

MSc in Biomedical Engineering 2018–2020
SP-LAB, Electronics Department, Constantine 1 University, Constantine, Algeria
Link: https://www.umc.edu.dz/index.php/en

BSc in Biomedical Engineering 2015–2018
Electronics Department, Constantine 1 University, Constantine, Algeria
Links: https://www.umc.edu.dz/index.php/en

PhD DISSERTATION

Dissertation Title: ***Medical Image Analysis with Deep Learning***
Supervisor: Prof. Said Benierbah
Rank in the PhD admission exam: **1st / 127**
Defense date: January 09, 2025
Distinction: **Very honorable**
This dissertation presented novel deep learning models for the automatic denoising of LDCT images and the automatic localization of ICH in CT images.

MASTER'S THESIS

Thesis Title: ***Classification and Segmentation of COVID-19 CXR and Chest CT Images using Deep Learning Algorithms***
Supervisor: Prof. Said Benierbah

Rank: **1st / 33**
This thesis proposed advanced deep learning algorithms for the automatic classification and segmentation of COVID-19 in CXR and CT images.

RESEARCH FUNDING & GRANT WRITING

• **Marie Skłodowska-Curie Postdoctoral Fellowship (MSCA-PF):** Lead applicant and proposal author.

ACADEMIC SERVICES

Reviewing Editor, Springer Nature Reviewer Communities (2025–present)
Peer-review activity: 45 peer reviews recorded on ORCID across 14 journals.
Conference Reviewer: *MICCAI*, *MIDL*, *ISBI*, *BMVC*
Journal Reviewer: *Medical Image Analysis*, *Computers in biology and medicine*, *Artificial Intelligence Review*, *Neurocomputing*, *Applied Intelligence*, *Biomedical Signal Processing and Control*, *Pattern Recognition Letters*, etc.

RESEARCH EVALUATION ACTIVITIES

Grant Proposal Evaluator for the 2026 EIG CONCERT-Japan Joint Call on AI-Powered Robotics for Real-World Applications

TECHNICAL SKILLS

Programming: Python, Matlab
Frameworks: PyTorch, TensorFlow, OpenCV, Scikit-learn, and others
Tools: NumPy, Pandas, HDF5, LaTeX
Systems: Linux, Windows

ADDITIONAL

MSc Thesis Examiner: Marvin Üürike (January 2026). Foundations for Predictive Analytics in Supercapacitor Manufacturing. University of Tartu.
Languages: English & French (professional-level proficiency)

LIST OF PUBLICATIONS

Journal Articles: 4 (all as the first author)

1. **Abdesselam Ferdi**, Abdelmalik Taleb-Ahmed, Amir Nakib, and Youcef Ferdi. Electrostatic Force Regularization for Neural Structured Pruning. *Information Sciences*. 2026.
Link: https://www.sciencedirect.com/science/article/pii/S002002552501182X
2. **Abdesselam Ferdi**. Lightweight G-YOLOv11: Advancing Efficient Fracture Detection in Pediatric Wrist X-rays. *Biomedical Signal Processing and Control*. 2026.

Link: https://www.sciencedirect.com/science/article/abs/pii/S1746809425013722

3. **Abdesselam Ferdi**, Said Benierbah, and Amir Nakib. Residual encoder-decoder based architecture for medical image denoising. *Multimedia Tools and Applications*. 2025.
Link: https://link.springer.com/article/10.1007/s11042-024-20175-1

4. **Abdesselam Ferdi**, Said Benierbah, Amir Nakib, Youcef Ferdi, and Abdelmalik Taleb-Ahmed. Quadratic Convolution-based YOLOv8 (Q-YOLOv8) for localization of intracranial hemorrhage from head CT images. *Biomedical Signal Processing and Control*. 2024.
Link: https://www.sciencedirect.com/science/article/abs/pii/S1746809424006694

Conference articles: 6 (all as the first author)

1. **Abdesselam Ferdi**. Deep Convolutional Neural Networks Structured Pruning via Gravity Regularization. *IEEE International Conference on Machine Learning & Applied Network Technologies*. 2024.
Link: https://ieeexplore.ieee.org/document/10908799

2. **Abdesselam Ferdi** and Amir Nakib. YOLO based Object detection Enhancement based on Quadratic Convolution. *International Conference on Foundation and Large Language Models*. 2024.
Link: https://ieeexplore.ieee.org/document/10852440

3. **Abdesselam Ferdi**, Said Benierbah, and Youcef Ferdi. YOLOv3-based Intracranial Hemorrhage Localization from CT Images. *International Symposium on Advanced Topics in Electrical Engineering*. 2023.
Link: https://ieeexplore.ieee.org/document/10108273

4. **Abdesselam Ferdi**, Said Benierbah, and Youcef Ferdi. Colorization Technique to improve DCNN-based ICH CT Image Classification. *Conférence Nationale sur les Télécommunications et ses Applications*. 2023.

5. **Abdesselam Ferdi**, Said Benierbah, and Youcef Ferdi. U-Net-based Semantic Segmentation of COVID-19 Chest CT Images: A Transfer Learning Approach. *International Conference on Image and Signal Processing and their Applications*. 2022.
Link: https://ieeexplore.ieee.org/document/9786341

6. **Abdesselam Ferdi**, Said Benierbah, and Youcef Ferdi. Five Deep Learning Models for Classification of Chest X-Ray Images of COVID-19. *International Conference on Electronics, Artificial Intelligence and New Technologies*. 2021.

Under Review: 1 (as the first author)

1. **Abdesselam Ferdi** and Radwa El Shawi. Language Optimization Concept Bottleneck Models. Under Review. 2026.